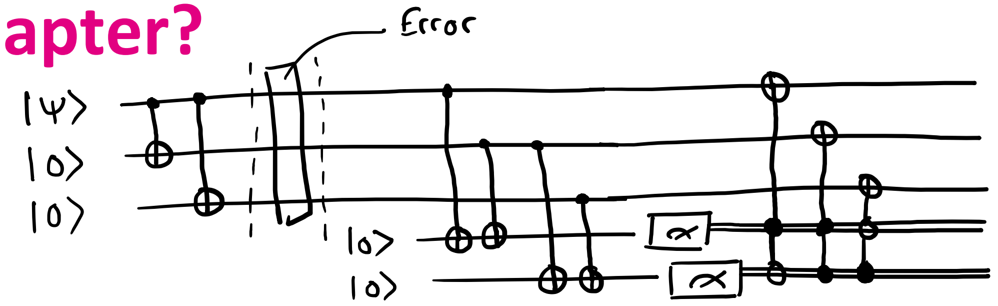
Pinakin Padalia

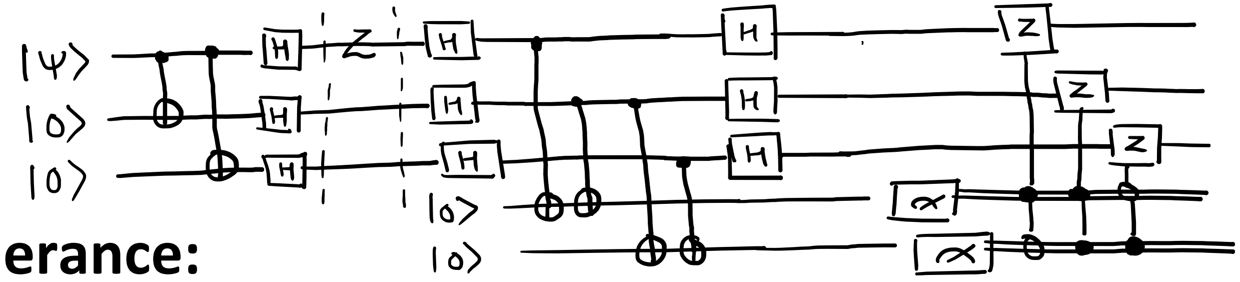
Quantum Error Mitigation(III)

Shor Code and Fault Tolerance

Pinakin Padalia

What are we covering in this chapter?





- Bit Flip Error Mitigation
- Phase Flip Error Mitigation

- **Shor code (9 qubit code) and fault tolerance:**
  - Strategy for error mitigation: Recap
  - Shor Code
  - Fault tolerance

Strategy for error mitigation: Recap

- Strategy:

$|\psi\rangle$ — $\boxed{\text{Encode}}$ — {Error} — $\boxed{\text{Detect}}$ — $\boxed{\text{Correct.}}$ —

Encode: More qubits.

Error: Done multiple times.

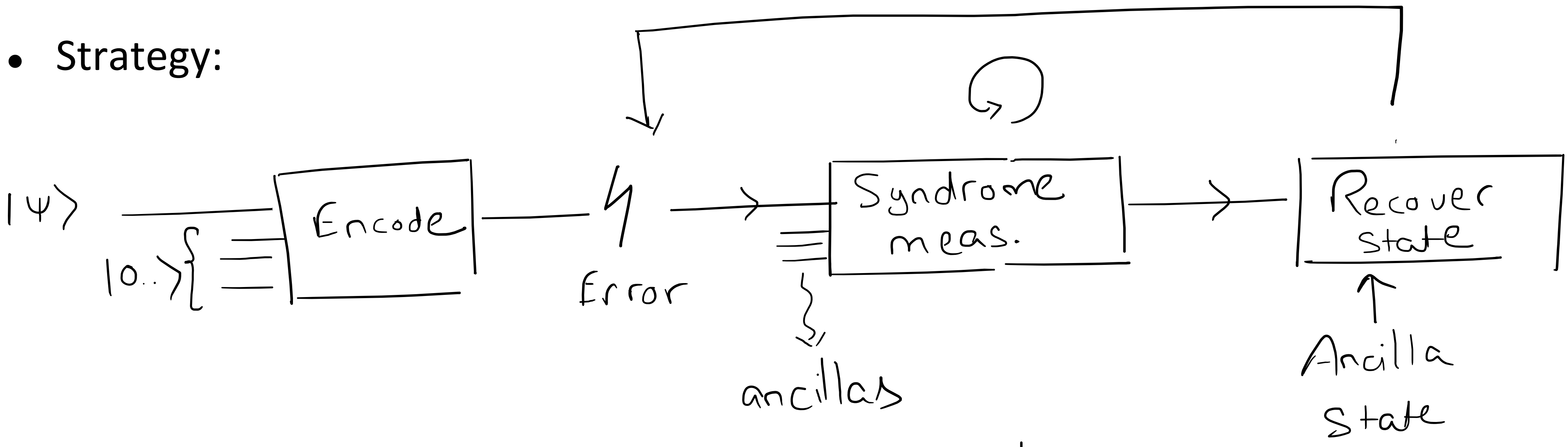
Detect: Using Ancillas

forces errors to be discrete.

Correct. OR Recover Based on Ancilla Outputs.

Strategy for error mitigation: Recap

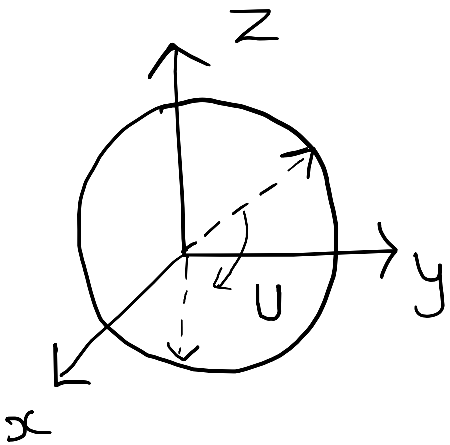
- Strategy:



Measurement done on ancillas

Shor Code

- Error can be:
  - Bit flip (Pauli X)
  - Phase flip (Pauli Z)
  - Or Both (Pauli ZX = iY)



- A general unitary error on a single qubit:

Linear Combination

$$U = \alpha I + \beta X + \gamma Z + \delta Y$$

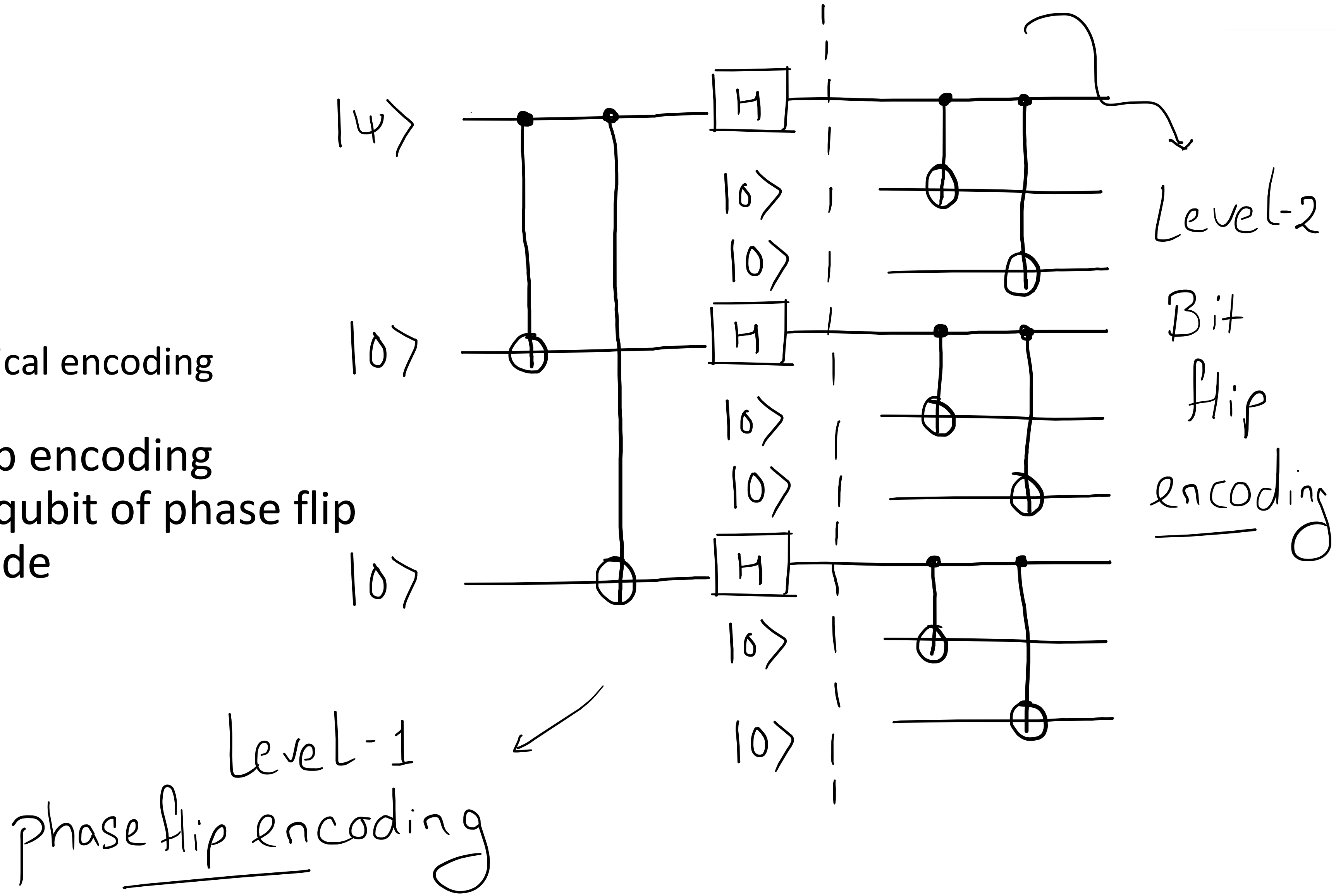
Coefficients

- Shor Code:
  - Arbitrary single qubit error correction
  - As long as the errors are unitary
  - Combines the bit flip and phase flip correction codes

Shor Code

- Shor Code
- Mitigation:
  - Step 1: Encoding
  - Idea: Use hierarchical encoding

- Start with phase flip encoding
- Then encode each qubit of phase flip code into bit flip code



Shor Code

- Shor Code
- Mitigation:
  - Step 1: Encoding
  - Idea: Use hierarchical encoding

- Encoded state:

$$|0\rangle_L \xrightarrow{\text{level 1}} |+++\rangle \xrightarrow{\text{level 2}}$$

$$\left(\frac{|000\rangle+|111\rangle}{\sqrt{2}}\right)\left(\frac{|000\rangle+|111\rangle}{\sqrt{2}}\right)\left(\frac{|000\rangle+|111\rangle}{\sqrt{2}}\right)$$

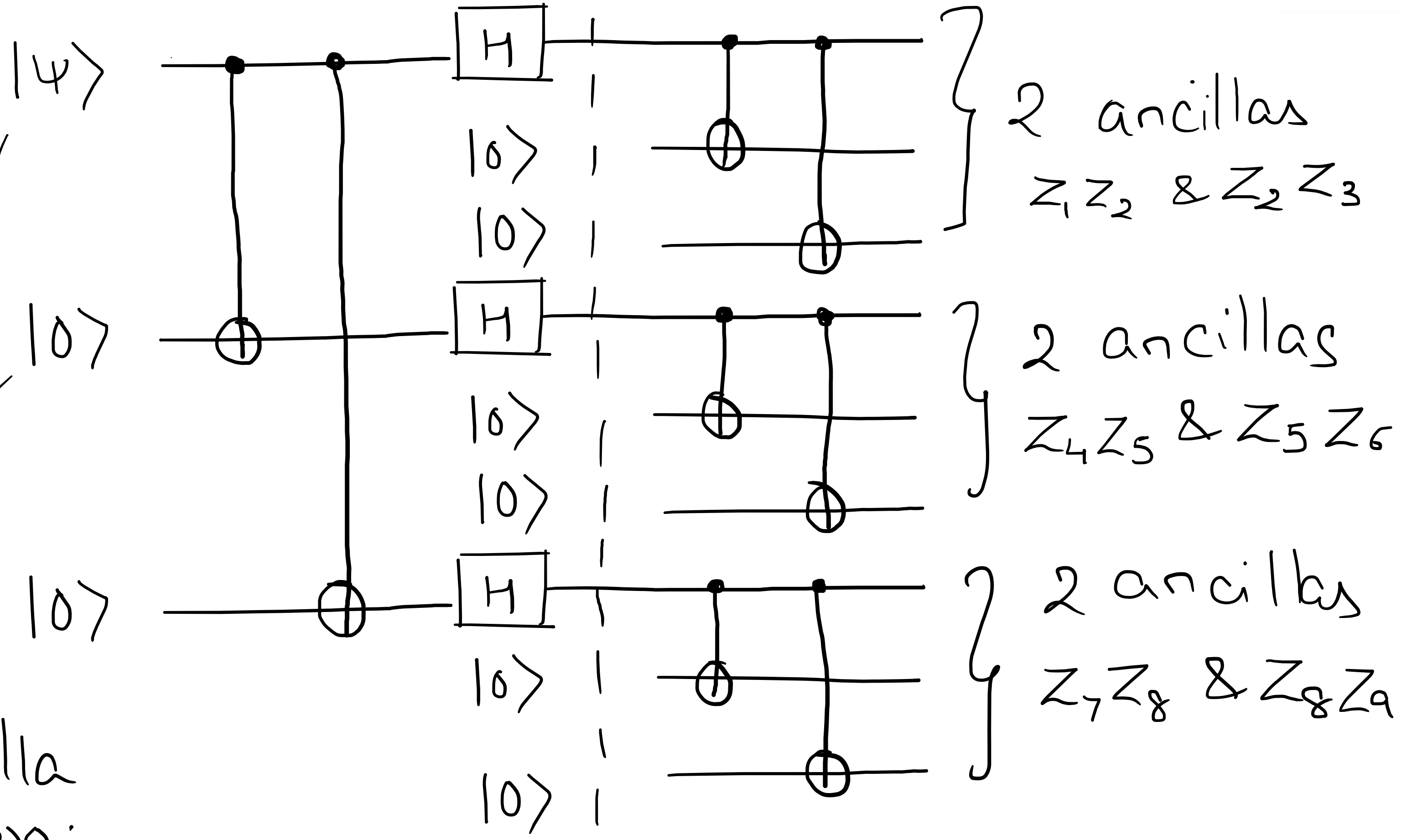
$$|1\rangle_L \xrightarrow{\text{level 1}} |---\rangle \xrightarrow{\text{level 2}}$$

$$\left(\frac{|000\rangle-|111\rangle}{\sqrt{2}}\right)\left(\frac{|000\rangle-|111\rangle}{\sqrt{2}}\right)\left(\frac{|000\rangle-|111\rangle}{\sqrt{2}}\right)$$

Shor Code

Total 8 ancillas

- Shor Code
- Mitigation:
  - Step 2: Adding ancillas



1 ancilla for checking phase flip b/w these 2 groups $X_1X_2X_3X_4X_5X_6$

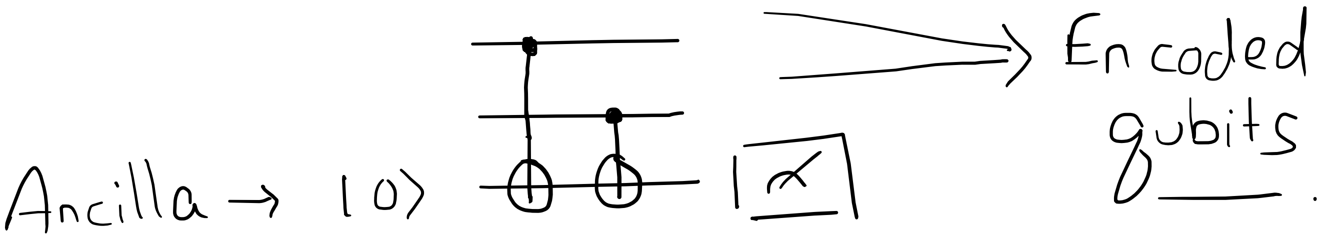
1 ancilla here too: $X_4X_5X_6X_7X_8X_9$

Shor Code

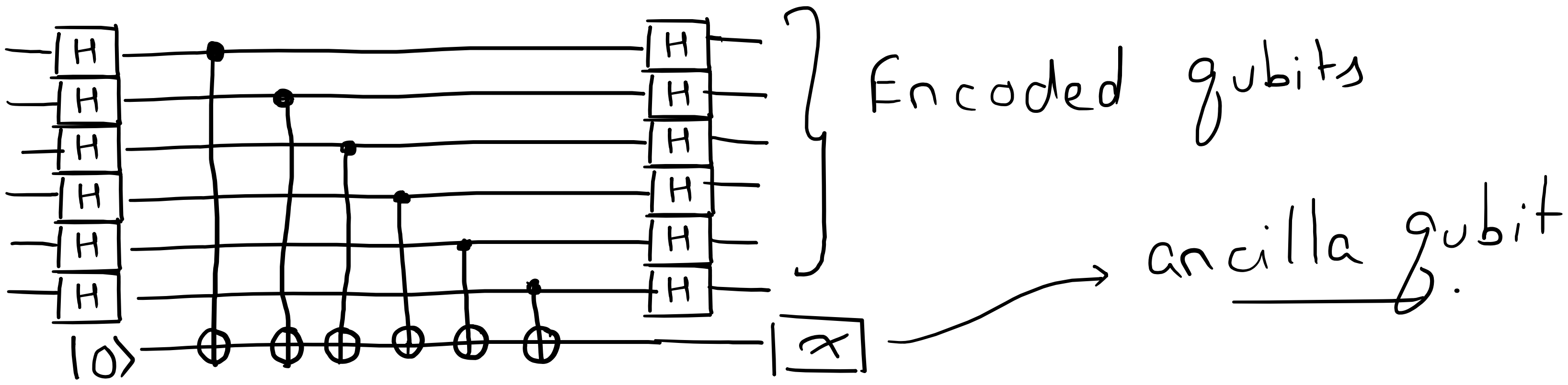
Total 8 ancillas.

- Shor Code
- Mitigation:
  - Step 2: Adding ancillas

$Z_x Z_y$ ancilla structure:

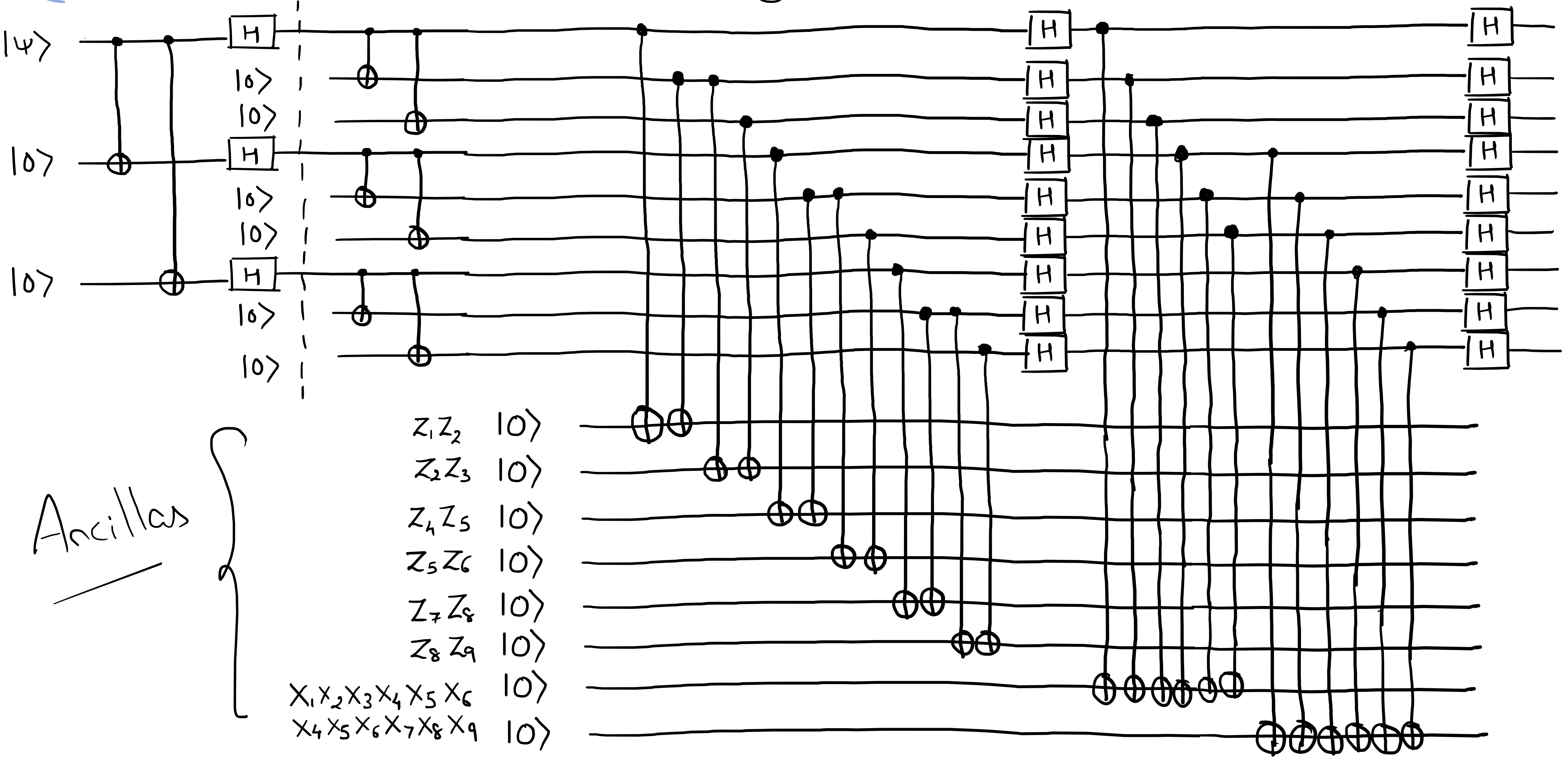


$X_1 X_2 X_3 X_4 X_5 X_6$ ancilla structure:



Shor Code

Single Logical Qubit !



Shor Code

- Shor Code
- Mitigation:
  - Step 3: Syndrome Measurement

Similar to Bit flip code

for $Z_x Z_y$

& Similar to phase flip code

for $X_1 \dots X_6$

Example:

$Z_1 Z_2 = 0$ & $Z_2 Z_3 = 1$ $\Rightarrow$ Bit flip in 3

$X_1 X_2 X_3 X_4 X_5 X_6 = 0$ &
$X_4 X_5 X_6 X_7 X_8 X_9 = 1$
} Phase flip in last group

Shor Code

- Shor Code
- Mitigation:
    - Step 4: Recovery

Similar to Bit flip code

for $Z_x Z_y \longrightarrow$ Apply X corr.

& Similar to phase flip code

for $X_1 \ldots X_6 \longrightarrow$ Apply Z corr.

Example:

$Z_1 Z_2 = 0$ & $Z_2 Z_3 = 1 \Rightarrow$ Bit flip in 3 $\longrightarrow$ Pauli X on 3

$X_1 X_2 X_3 X_4 X_5 X_6 = 0$ &
$X_4 X_5 X_6 X_7 X_8 X_9 = 1$

$\left.\begin{array}{l}\text{Phase flip} \\ \text{in last group}\end{array}\right\}$ Pauli Z on 7, 8, 9

Shor Code

- Shor Code
  - Multiple single qubit bit flip errors can be corrected, if they appear in different clusters
  - Errors can be corrected just like bit and phase flip codes even without doing measurements with the ancillas
  - Not an optimal code – a degenerate code
  - Total qubit requirements: 9 for encoding and 8 ancillas = 17 qubits for 1 logical qubit!
- Other encoding schemes and error correction methods:
  - 5 Qubit code
  - Stene's 7 qubit code
  - Surface codes

Fault Tolerance

- Fault Tolerance:
  - With physical error rates below a certain threshold, one can make the logical qubit errors to be arbitrarily low using error correction
  - A logical qubit will be error resistant even if physical qubits are poor in quality
  - Cost: Error correction and requirement of more physical qubits

- Requirements:
  - Perform error detection and correction without measurements
  - Perform error correction for the errors introduced due to gates

Thank you for your attention!

References

- Michael A. Nielsen and Isaac L. Chuang. 2011. Quantum Computation and Quantum Information: 10th Anniversary Edition (10th. ed.). Cambridge University Press, USA.
- M. Wilde, "From Classical to Quantum Shannon Theory", Arxiv.org, 2019. [Online]. Available: https://arxiv.org/pdf/1106.1445.pdf.
- J. Preskill, "Lecture Notes for Physics 229: Quantum Information and Computation", Lorentz.leidenuniv.nl, 1998. [Online]. Available: https://www.lorentz.leidenuniv.nl/quantumcomputers/literature/preskill_1_to_6.pdf.